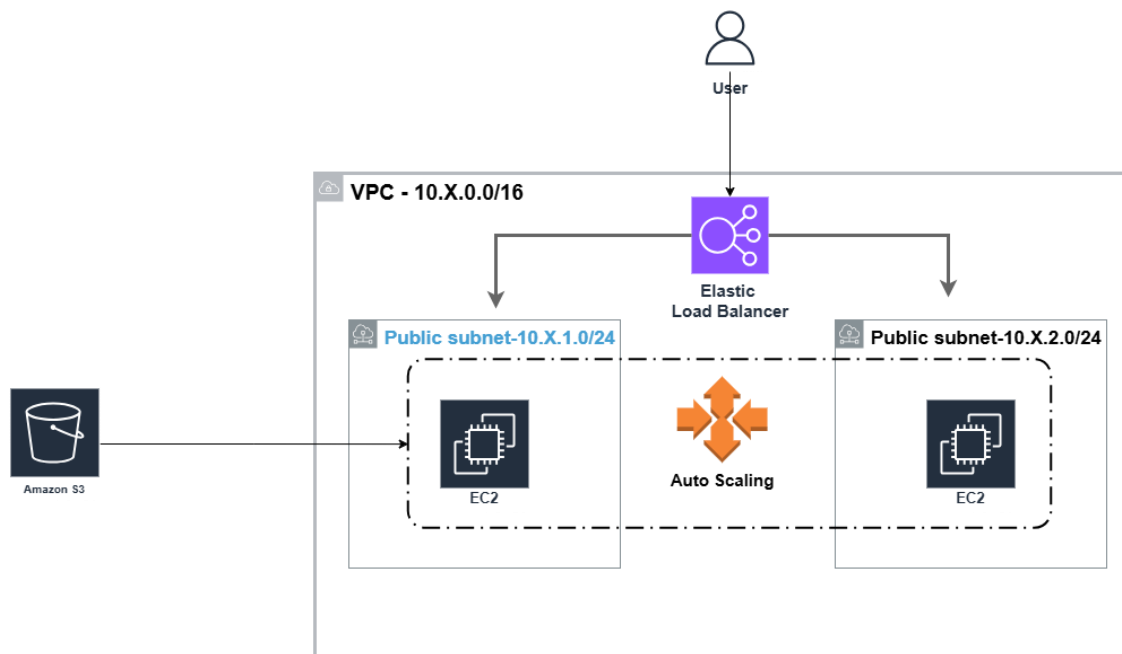


1a. E-Commerce Website Deployment

A startup company is launching a new e-commerce website. The application must handle increasing customer traffic, especially during sales events and festivals. The company wants a highly available and scalable architecture on AWS so that the website remains accessible even if traffic increases or one server fails. The system should also store product images, user data, and order information securely.

Tasks

- Create an IAM user & assign necessary permissions.
- Deploy the E-commerce application on EC2 instances.
- Traffic distributed between EC2 instances.
- Instances scale automatically during high load.
- Store product images in s3 bucket.



1b. Write a shell script to display the message:

"Welcome to DevOps Learning"

1c. Write a Python program to remove duplicates from a sorted list and return the new length.

1d. Create a repository in GitHub and push shell script and python code to the main branch.

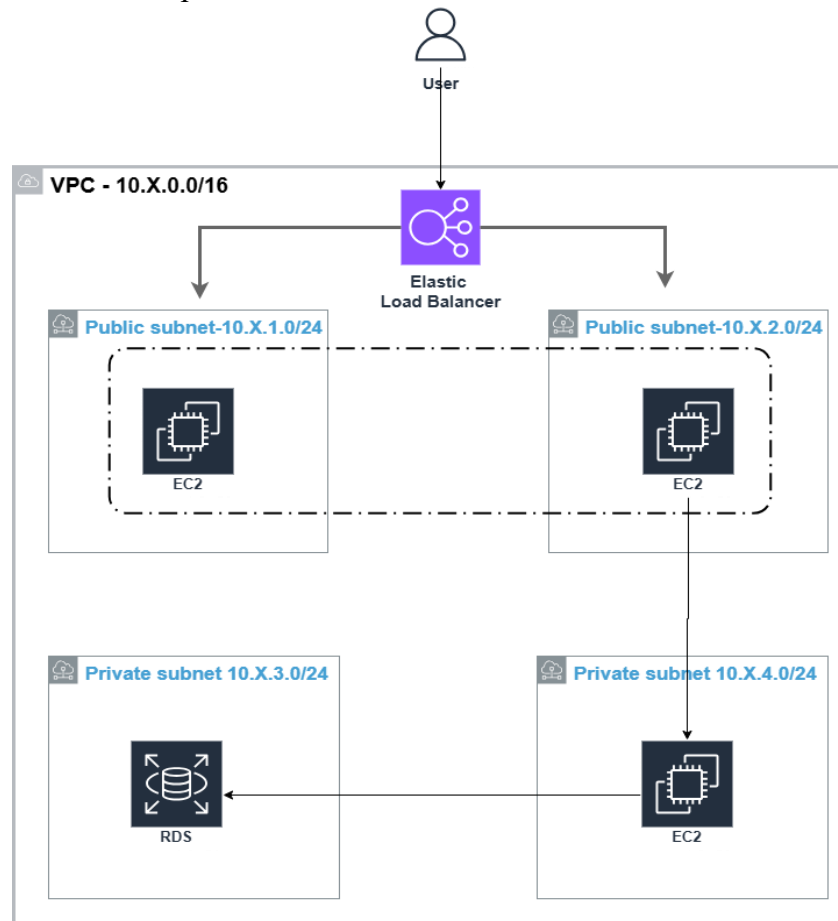
2a. Hospital Management System

A hospital wants to deploy a cloud-based patient management system where doctors can access patient records and medical reports.

The system must store medical records securely and allow access only to authorized users.

Tasks

- Create an IAM user & assign necessary permissions.
- Create a VPC with public and private subnets.
- Deploy the hospital application on EC2 instances.
- Traffic distributed between EC2 instances.
- Create an RDS database for patient records.



2b. Create a script that prints:

- Current logged-in user
- Hostname
- Present working directory

2c. Write a program to remove all occurrences of a given value from a list and return the updated list length.

2d. Create a repository in GitHub and push shell script and python code to the main branch.

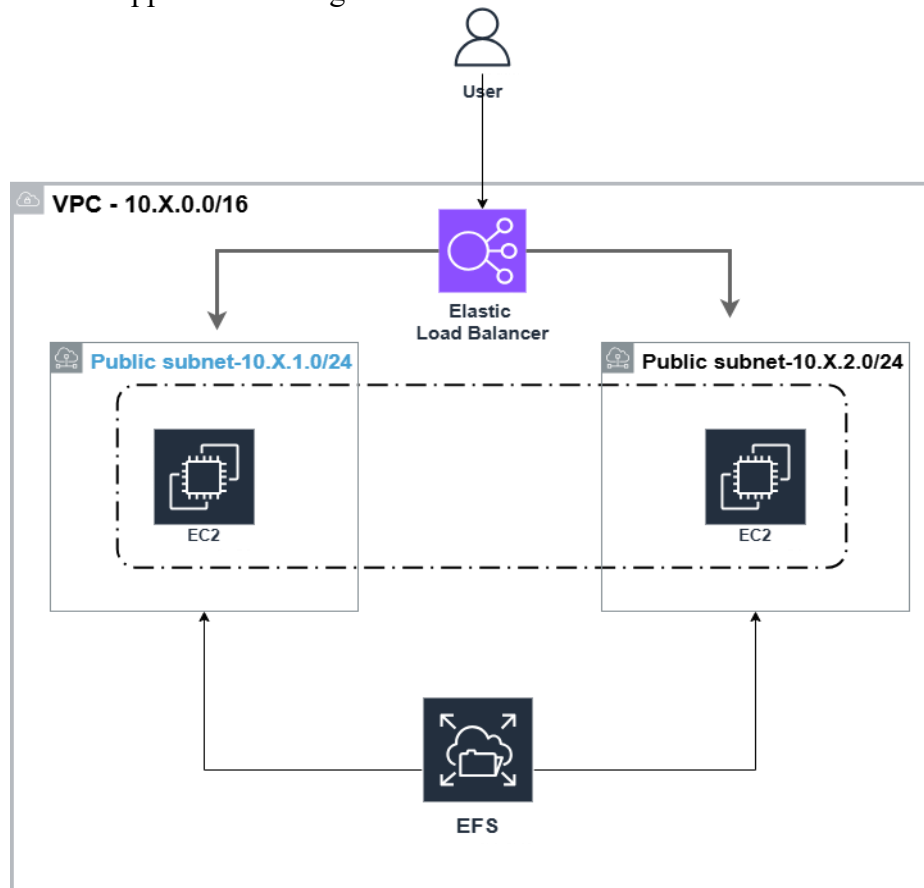
3a. Video Streaming Platform

A company wants to build a video streaming platform similar to Netflix.

The platform must support millions of users watching videos simultaneously and store large video files.

Tasks

- Create an IAM user & assign necessary permissions.
- Launch EC2 instances to host the streaming application.
- Traffic distributed between EC2 instances.
- Use EFS for shared application storage.



3b. Write a script that:

- Takes user name as input
- Displays: "Hello <name>, welcome!"

3c. Write a program to find the index where a target element should be inserted in a sorted list.

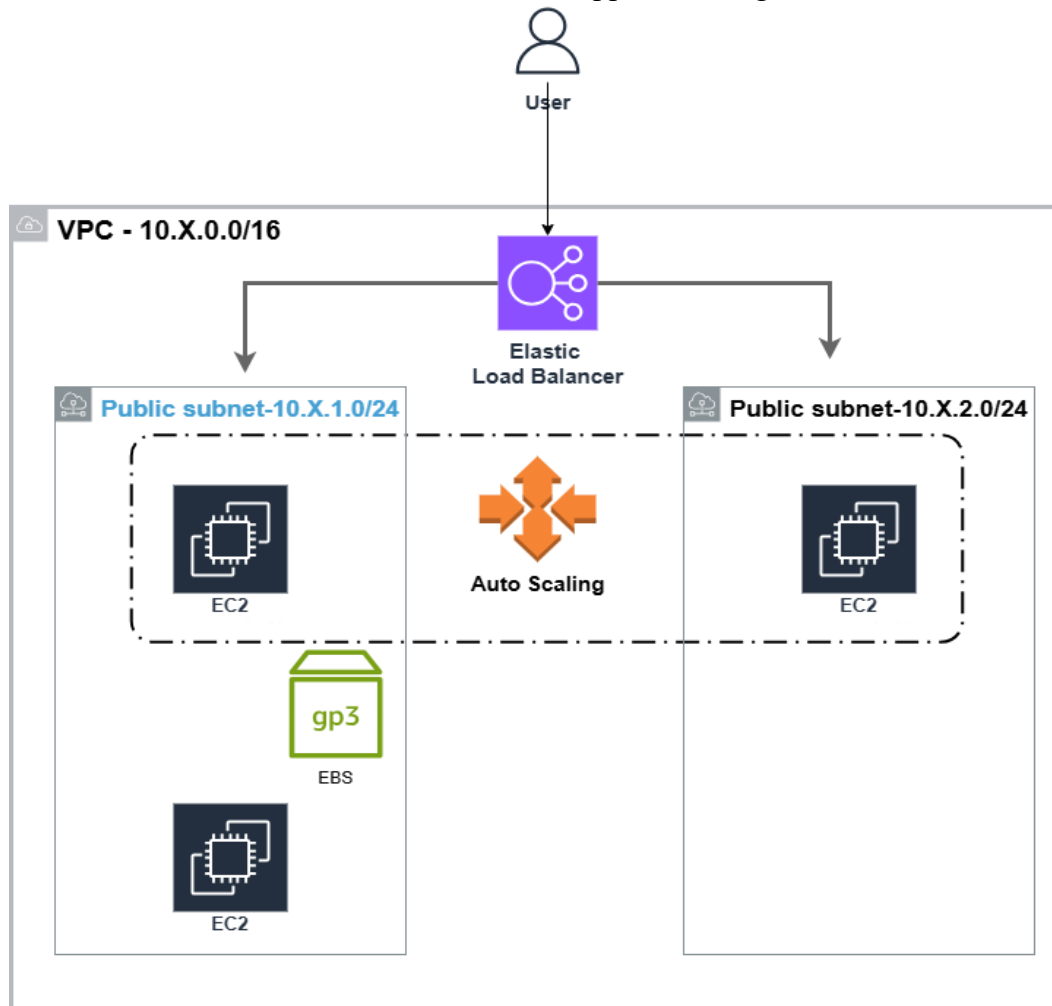
3d. Create a repository in GitHub and push shell script and python code to the main branch.

4a. Startup Web Application

A startup company has developed a web application that is expected to grow rapidly. They need a scalable cloud infrastructure to handle increasing traffic.

Tasks

- Create an IAM user & assign necessary permissions.
- Launch EC2 instances for application hosting.
- Configure Application Load Balancer.
- Create an Auto Scaling Group.
- Attach EBS volumes to each EC2 instance to store application logs and transaction data.



4b. Create a script that:

- Accepts two numbers from user
- Prints their sum

4c. Write a Python program to find the contiguous subarray with the largest sum.

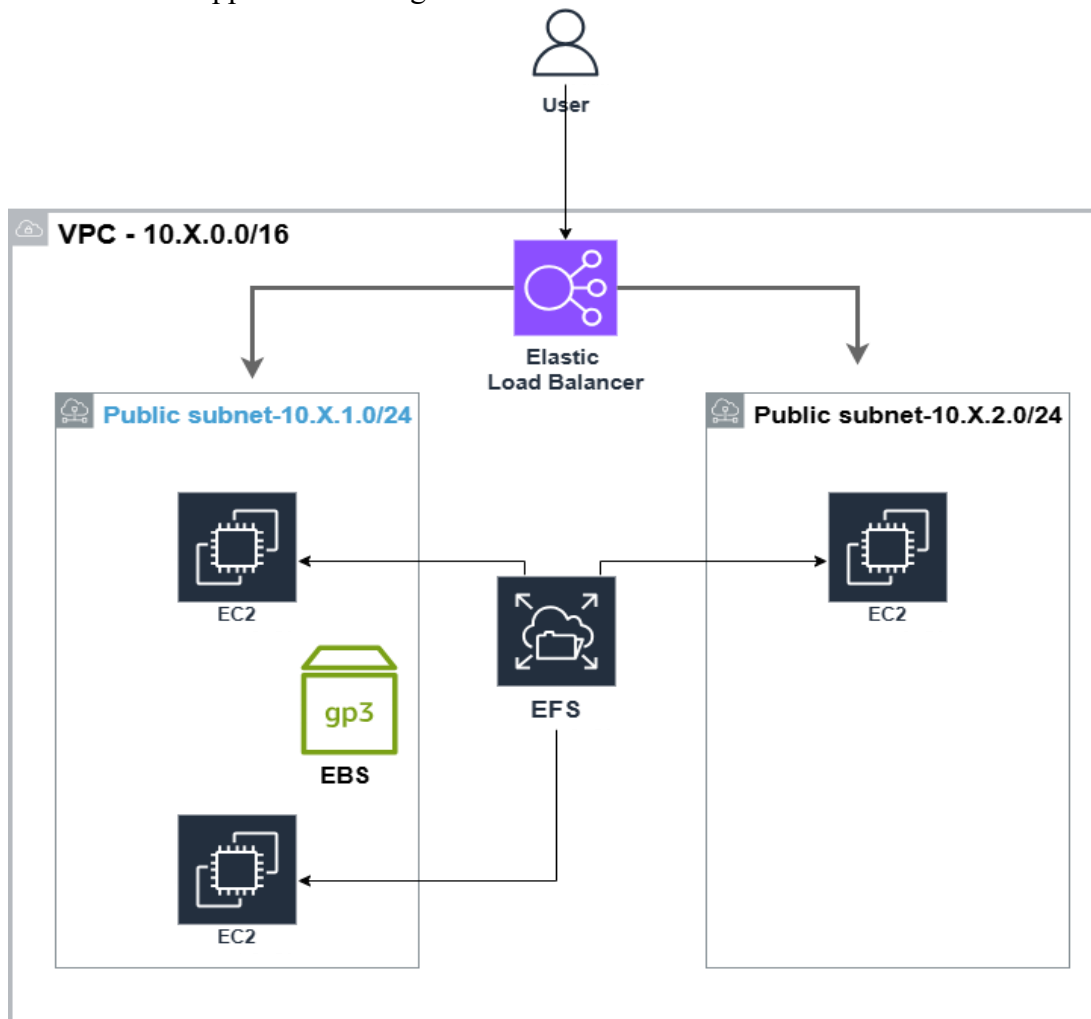
4d. Create a repository in GitHub and push shell script and python code to the main branch.

5a. Banking Application

A bank wants to deploy an online banking application where customers can check balances and perform transactions. The system must be secure and highly available.

Tasks

- Create an IAM user & assign necessary permissions.
- Launch two EC2 instances in the public subnet.
- Install a web server (Apache/Nginx) on EC2.
- Attach EBS volumes to each EC2 instance to store application logs and transaction data.
- Create an Application Load Balancer to distribute traffic.
- Use EFS for shared application storage



5b. Write a script to:

- Accept a number
- Check whether it is even or odd

5c. Write a program to find the length of the last word in a given string.

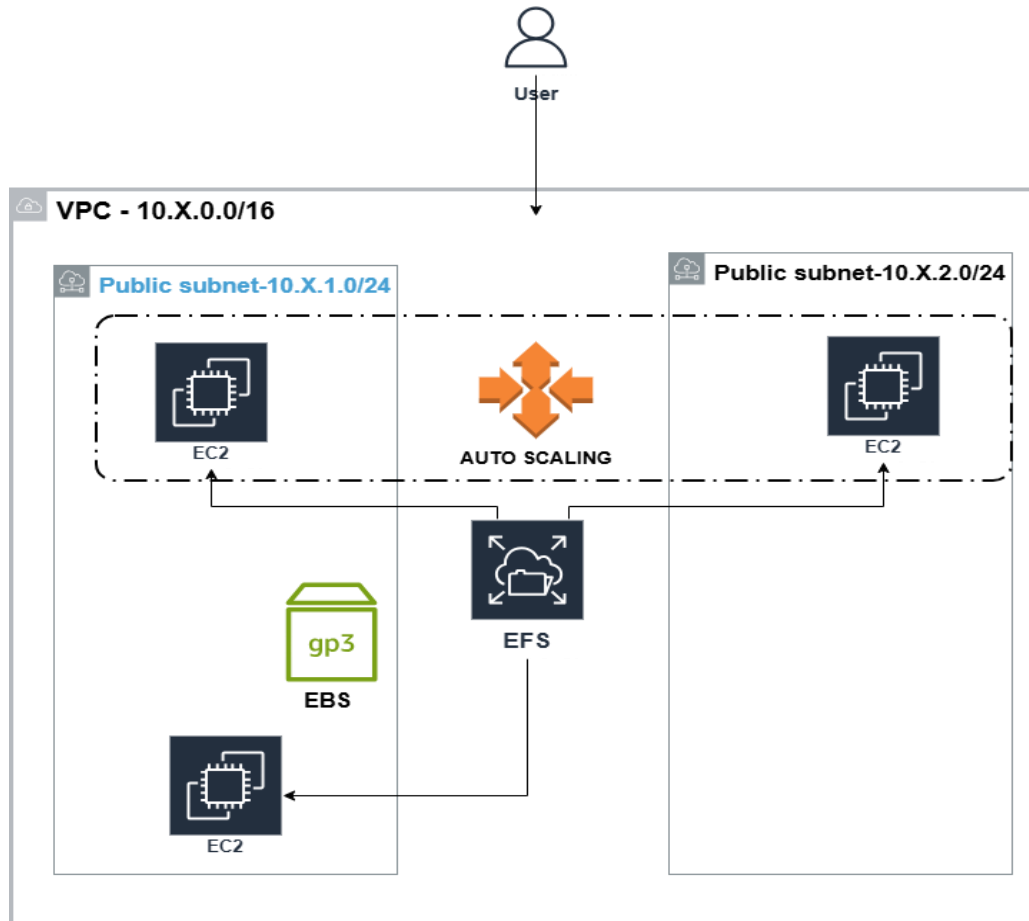
5d. Create a repository in GitHub and push shell script and python code to the main branch.

6a. News Website

A media company wants to host a news website that receives high traffic during breaking news events.

Tasks

- Create an IAM user & assign necessary permissions.
- Launch two EC2 instances for the web application.
- Attach EBS volumes for storing news articles.
- Configure Auto Scaling to handle traffic spikes.
- Use EFS for shared application storage.



6b. Write a script that:

Takes a path as input

Checks whether it is:

- File
- Directory
- Or does not exist

6c. Write a program that takes a list representing a number and adds one to it.

Example: [1,2,3] → [1,2,4]

6d. Create a repository in GitHub and push shell script and python code to the main branch.

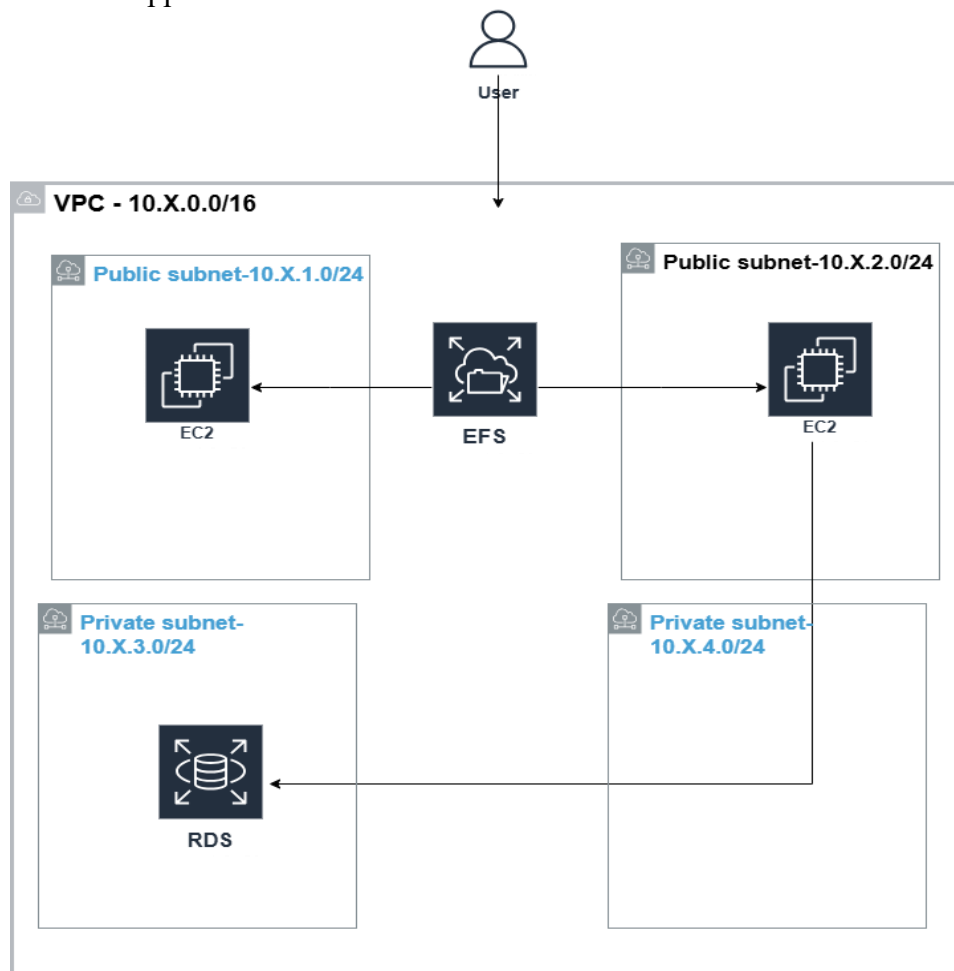
7a. Food Delivery Application

A startup wants to build an online food delivery application where users can browse restaurants, place orders, and track delivery status.

The system must support high traffic during lunch and dinner hours and ensure that order data is stored reliably.

Tasks

- Create an IAM user & assign necessary permissions.
- Create a VPC with public and private subnets.
- Launch EC2 instances to host the application.
- Deploy RDS database to store order details and customer information.
- Create EFS to share application files between servers.



7b. Write a script to:

- Print numbers from 1 to 10 using a for loop

7c. Write a program to add two binary strings and return the result as a binary string.

7d. Create a repository in GitHub and push shell script and python code to the main branch.

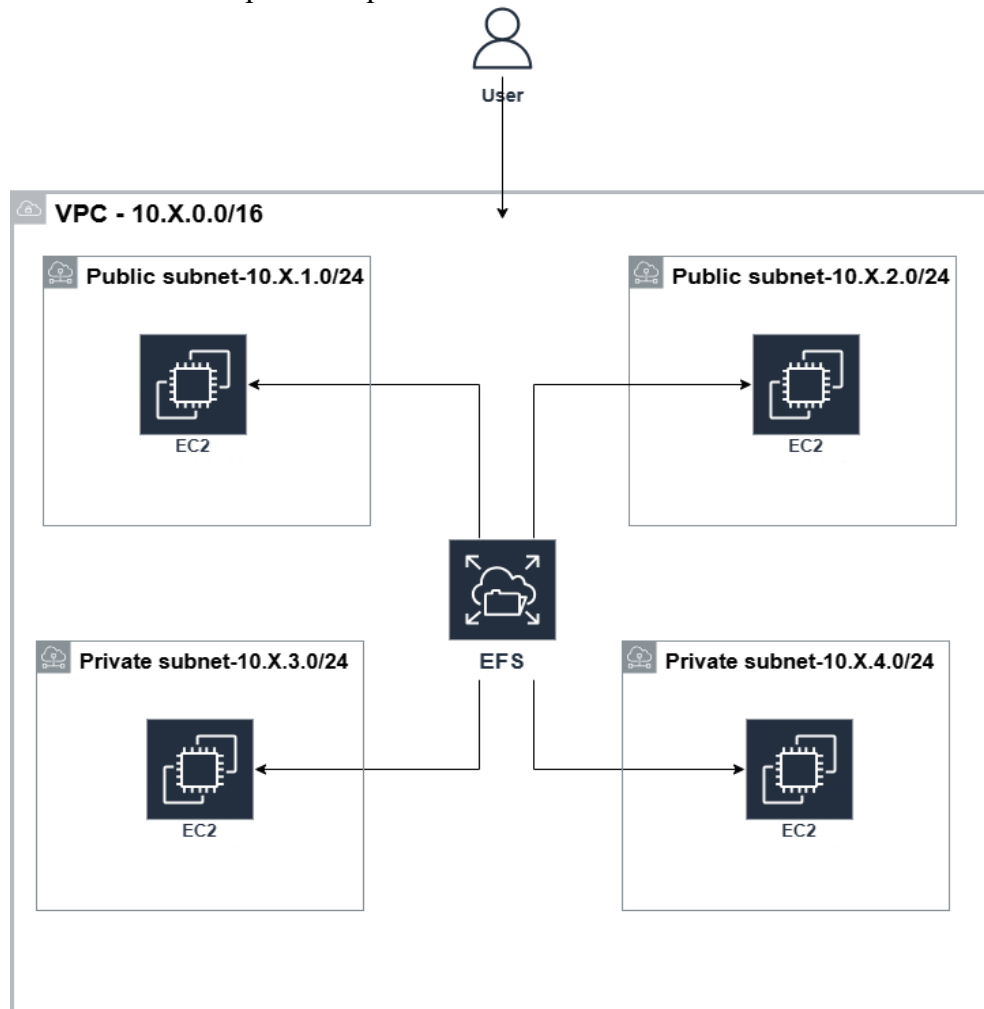
8a. Social Media Application

A company wants to launch a social media platform where users can upload photos, write posts, and interact with friends. The platform must support thousands of users simultaneously.

Uploaded photos and videos must be stored reliably, and user data must be stored in a database.

Tasks

- Create an IAM user & assign necessary permissions.
- Create a VPC with a public and private subnet.
- Launch EC2 instances to host the social media application.
- Create EFS and attach with public & private servers.



8b. Write a script to:

- Print numbers from 10 to 1 using a while loop

8c. Write a Python function to compute the integer square root of a number without using built-in functions.

8d. Create a repository in GitHub and push shell script and python code to the main branch.

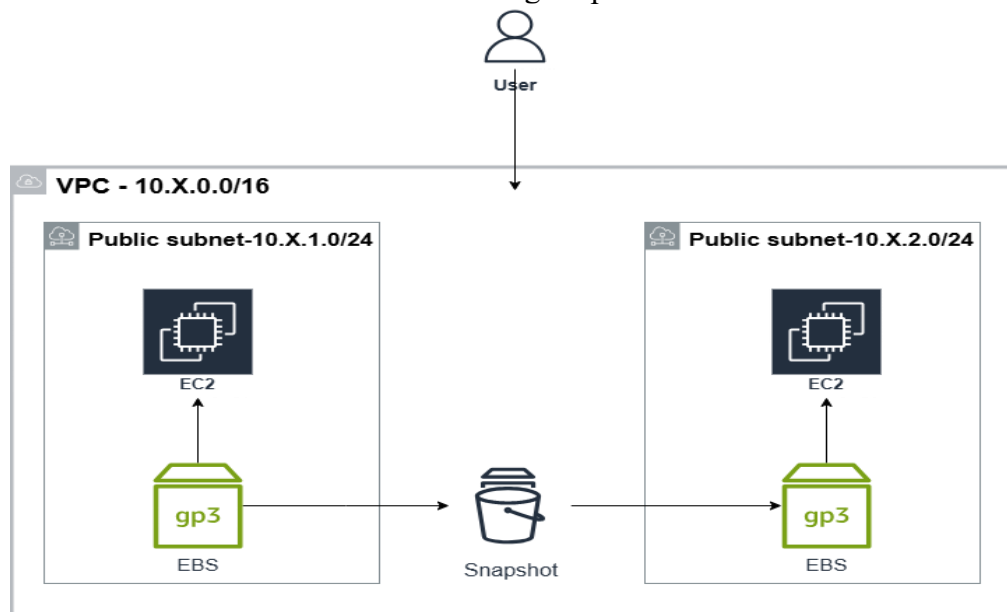
9a. Online Ticket Booking System

A travel company wants to deploy an online ticket booking platform where users can book bus, train, or flight tickets.

Booking confirmations and tickets must be stored securely.

Tasks

- Create an IAM user & assign necessary permissions.
- Create a VPC with public subnets.
- Launch two EC2 instances for the ticket booking application.
- Attach EBS volumes to store booking logs.
- Move the EBS volume from AZ-A to AZ-B using snapshot.



9b. Create a script that:

- Displays disk usage of the system
- Filters only the root (/) partition

9c. Write a program to find how many distinct ways you can climb n stairs if you can take 1 or 2 steps at a time.

9d. Create a repository in GitHub and push shell script and python code to the main branch.

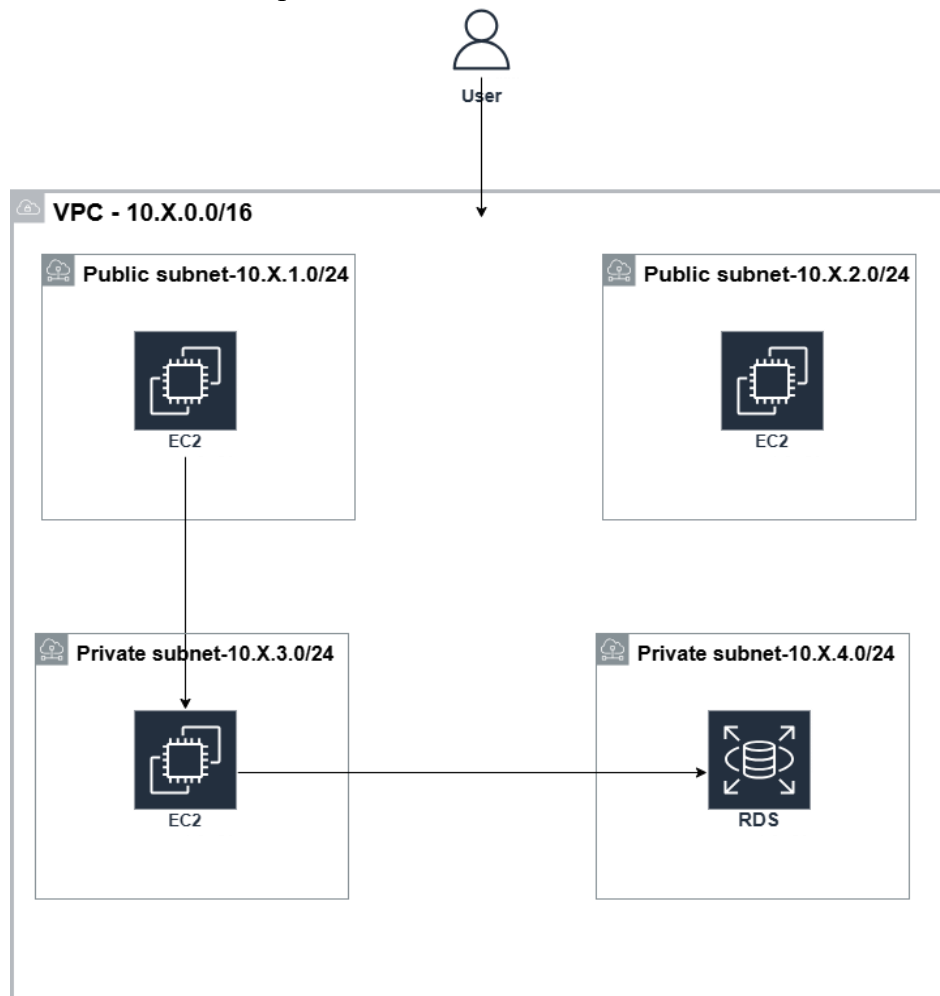
10a. Corporate Document Management System

A large company wants to build a document management system where employees can upload and access company documents such as policies, reports, and presentations.

Documents must be stored securely, and multiple servers should be able to access shared files.

Tasks

- Create an IAM user & assign necessary permissions.
- Create a VPC with public and private subnets.
- Launch EC2 instances to host the document management application.
- Create RDS and connect with a private server.



10b. Write a script that:

- Takes directory path as input
- Lists all files inside it
- Shows an error if directory does not exist

10c. Write a program to merge two sorted lists into one sorted list.

10d. Create a repository in GitHub and push shell script and python code to the main branch.